

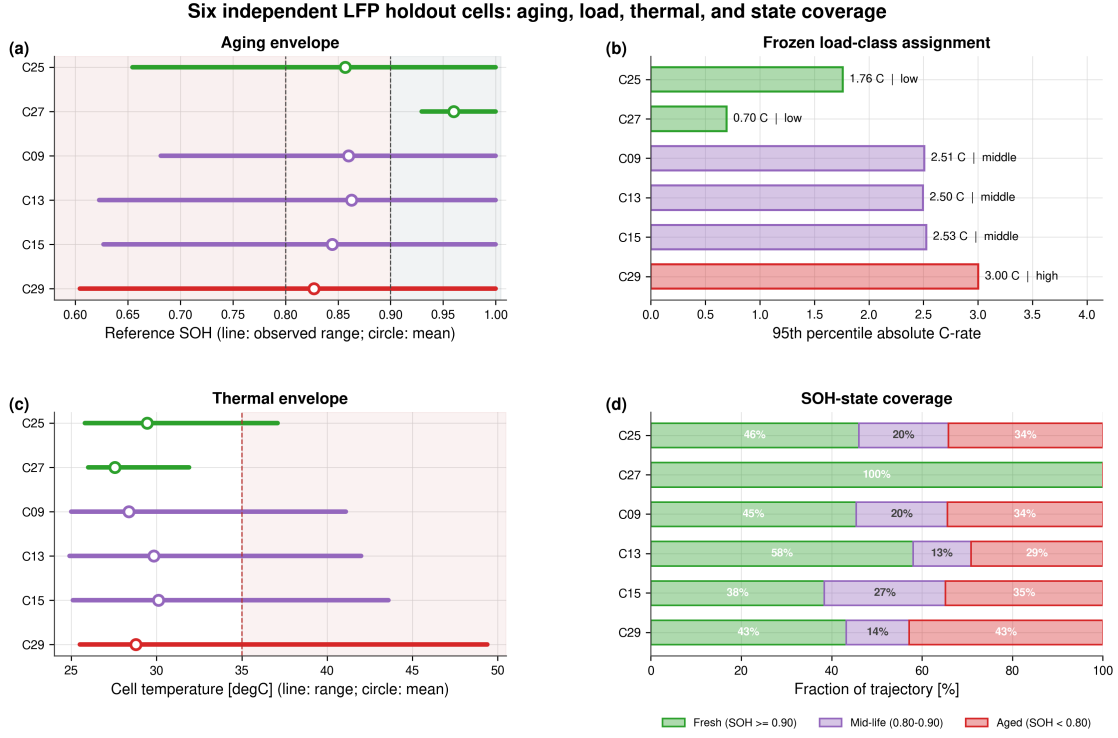


Figure 4.5: Measured coverage of the six-cell independent holdout test set. The panels show observed SOH, the descriptive load-class assignment, the thermal envelope, and full-trajectory SOH-state coverage. The high-load group contains one exploratory trajectory, while the low- and middle-load groups contain multiple cells.

4.1.3 SOC and SOH Reference-Label Construction

The LFP dataset contains measurements from which the labels for Chapters 6 and 7 are constructed. For a sample interval Δt_k , the signed transferred charge is

$$Q_k = \frac{I_k \Delta t_k}{3600}, \quad (4.2)$$

and the measured capacity of a dedicated capacity-test interval $\mathcal{T}_{\text{cap}}$ is

$$C_k = \sum_{j \in \mathcal{T}_{\text{cap}}} |Q_j|. \quad (4.3)$$

The embedded-AI study uses the conventional nominal-capacity definition

$$\text{SOH}_k = \frac{C_k}{C_{\text{nom}}}, \quad (4.4)$$